

## Berry-dipole semimetals

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We introduce “Berry-dipole semimetals”, whose band degeneracies are characterized by quantized Berry dipoles. Through a two-band model constructed by a Hopf map, we reveal that the Berry-dipole semimetals display a multitude of salient properties distinct from other topological semimetals. On the boundary, we find that the first-order Berry-dipole semimetal harbors anomalous paired Fermi arcs with the same spin polarization, even though the layer Chern number is zero, and the second-order Berry-dipole semimetal hosts dispersionless hinge arcs. In the bulk, we find that the low-energy Berry-dipole Hamiltonian near the band node has a quadratic energy dispersion and peculiar Berry curvature, which give rise to rather unique characteristics in the intrinsic anomalous Hall effect, orbital magnetization and Landau levels. Our study shows that Berry-dipole semimetals are a class of topological gapless phases supporting rich intriguing physics.

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Over the past decade, topological semimetals in three dimensions, including nodal-point and nodal-line semimetals, have been intensively and extensively investigated both in theory and in experiment [1–6]. The Weyl semimetal (WSM) is a paradigmatic nodal-point semimetal attracting particular interest [7–14]. The band degeneracy in WSMs, known as Weyl node (or point), is a singularity of the Berry curvature and a topological object whose stability does not need symmetry protection [15]. The low-energy Hamiltonian near the Weyl node, which resembles the relativistic Weyl Hamiltonian, gives rise to a linear energy dispersion and a Berry curvature structure whose momentum dependence is reminiscent of the field distribution of a magnetic monopole in real space [16,17]. Because of the relativistic linear dispersion and monopolelike Berry curvature, WSMs carry a diversity of peculiar properties, such as open Fermi arcs [7] and chiral Landau levels [18,19], and serve as a fertile playground to investigate exotic electromagnetic field responses [20–26] and interaction-driven phases [27–29].

Since the Weyl node acts as a charge of Berry flux, a natural question is whether one can extend the concept of charge to dipole. Naively, when there are two Weyl nodes with opposite charges, one may expect that they naturally consist of a Berry dipole. However, such a picture is rather trivial since, in WSMs, the Weyl nodes with opposite charges always show up in pairs [30–32]. More importantly, the physics remains dominated by the Weyl Hamiltonian, without the need of a dipole picture. Interestingly, a dipole can also be pointlike, and the local and global properties of a point dipole are completely different from a point charge. Therefore, an extension to point Berry dipole would be nontrivial.

A point dipole is formed by two opposite point charges that are infinitely close. If there is no symmetry protection,

however, the collision of two Weyl nodes with opposite charges will cause an annihilation of them. Recent works have shown that if two Weyl nodes are related by a mirror symmetry, their annihilation is avoided [33], and a point Berry dipole will form when they meet at the mirror plane. Furthermore, it was found that the critical point between two different three-dimensional (3D) crystalline Hopf insulators is also a point Berry dipole [34]. Inspired by the existence of point Berry dipole, in this work we introduce the concept of “Berry-dipole semimetal” (BDSM). Rather than as the critical point of two 3D crystalline Hopf insulators, we find that, using the dimension-reduction picture, the Berry-dipole node can also show up at the plane separating 2D Hopf-insulator/second-order-topological-insulator planes and normal-insulator planes in the 3D Brillouin zone [see the illustration in Figs. 1(a) and 1(b)], just like that the Weyl node sits at the plane separating 2D Chern-insulator planes and normal-insulator planes [35]. With this observation in mind, we use the Hopf map to construct a two-band model that can realize both first-order and second-order BDSMs. As expected, we find that the boundary and bulk states of the BDSMs exhibit many salient properties distinct from other topological semimetals.

**General theory.** The Hopf map refers to a map from 3-sphere ( $S^3$ ) to 2-sphere ( $S^2$ ). Mathematically, this map can be described by a map from a two-component complex spinor to a three-component real vector. Denoting the complex spinor as  $\zeta = (\zeta_1, \zeta_2)^T$  and the real vector as  $\mathbf{d} = (d_x, d_y, d_z)$ , they are related by the formula,  $\mathbf{d} = \zeta^\dagger \boldsymbol{\sigma} \zeta$ , where  $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$  are the Pauli matrices. To be more explicit,

$$d_x = 2\text{Re}(\zeta_1^* \zeta_2), d_y = 2\text{Im}(\zeta_1^* \zeta_2), d_z = |\zeta_1|^2 - |\zeta_2|^2. \quad (1)$$

Using  $d_i$  to construct a two-band Hamiltonian,  $\mathcal{H}(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$ , it has been demonstrated that diverse distinctive topological phases can be realized by appropriately choosing the complex spinor. Notable examples include

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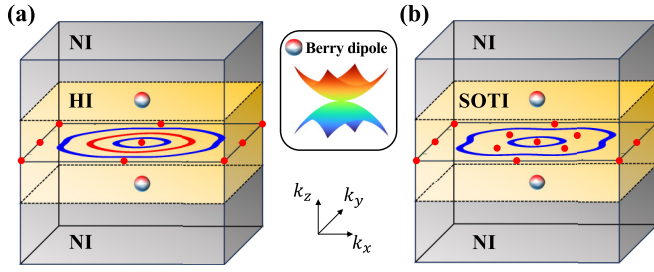


FIG. 1. Schematic of a first-order Berry-dipole semimetal (a) and a second-order Berry-dipole semimetal (b). The quadratic energy dispersion near the Berry-dipole node is sketched in the middle. In (a) and (b), the  $k_z$ -planes in yellow correspond to 2D Hopf insulators (HI) and second-order topological insulators (SOTI), respectively, and the  $k_z$  planes in gray correspond to normal insulators (NI). The blue and red lines (dots) refer to the solutions of  $d_z(\mathbf{k}) = 0$  and  $d_x(\mathbf{k}) = d_y(\mathbf{k}) = 0$  at the  $k_z = 0$  plane, respectively, which illustrate why the yellow planes are topologically nontrivial.

3D Hopf insulators [36–42], 3D nodal-link semimetals [43–45], 3D dipolar WSMs [46], 2D second-order topological insulators/superconductors [47–49], and Hopf defects harboring Majorana zero modes in 2D trivial superconductors [50]. In this work, we choose

$$\begin{aligned} \zeta_1(\mathbf{k}) &= \lambda(\sin k_x + i \sin k_y), \\ \zeta_2(\mathbf{k}) &= \delta(\cos k_x - \cos k_y) + i \left( M - t \sum_{i=x,y,z} \cos k_i \right). \end{aligned} \quad (2)$$

As will be shown below, the two-band model can realize both first-order and second-order BDSMs, depending on whether  $\delta$  is zero or finite.

*First-order BDSM.* We first consider the case with  $\delta = 0$ . The components of the Hamiltonian are then given by

$$\begin{aligned} d_x(\mathbf{k}) &= 2\lambda M(\mathbf{k}) \sin k_y, \quad d_y(\mathbf{k}) = 2\lambda M(\mathbf{k}) \sin k_x, \\ d_z(\mathbf{k}) &= \lambda^2 \sin^2 k_x + \lambda^2 \sin^2 k_y - M^2(\mathbf{k}). \end{aligned} \quad (3)$$

Here we have introduced  $M(\mathbf{k}) = (M - t \sum_{i=x,y,z} \cos k_i)$  to shorten the notation, and the lattice constants are set to unity throughout this work for notational simplicity. The energy spectra of the Hamiltonian are

$$E_{\pm}(\mathbf{k}) = \pm[\lambda^2(\sin^2 k_x + \sin^2 k_y) + M^2(\mathbf{k})]. \quad (4)$$

Apparently, the energy bands harbor nodes if  $M(\mathbf{k})$  has zeros on the four high symmetry  $k_z$  axes passing  $(k_x, k_y) = (0/\pi, 0/\pi)$ . Without loss of generality, we consider  $t < M < 3t$ , then there are two band nodes at  $\mathbf{k}_{n,\chi} = \chi(0, 0, \arccos(M/t - 2))$ , where  $\chi = \pm$ . To know the properties of the band nodes, we determine the low-energy Hamiltonians around the two band nodes, which read

$$\begin{aligned} \mathcal{H}_{\chi}(\mathbf{q}) &= 2\chi v v_z q_z q_y \sigma_x + 2\chi v v_z q_z q_x \sigma_y \\ &\quad + [v^2(q_x^2 + q_y^2) - v_z^2 q_z^2] \sigma_z, \end{aligned} \quad (5)$$

where  $v = \lambda$ ,  $v_z = t\sqrt{1 - (M/t - 2)^2}$  and  $\mathbf{q} = (q_x, q_y, q_z)$  denotes the momentum measured from the corresponding node. The low-energy spectra are

$$E_{\chi,\pm}(\mathbf{q}) = \pm[v^2(q_x^2 + q_y^2) + v_z^2 q_z^2]. \quad (6)$$

Obviously, the low-energy spectra are quadratic in all directions, which is distinct from the Weyl and Dirac semimetals hosting a linear dispersion near the band nodes.

A well-known characteristic of the Weyl node is that an integral of the Berry curvature over a closed surface enclosing it is quantized. The quantized value corresponds to the first-class Chern number and labels the topological charge of the Weyl node. For the low-energy Hamiltonians in Eq. (5), their Berry curvatures are given by [51]

$$\Omega_{\chi}^{(c)}(\mathbf{q}) = -\Omega_{\chi}^{(v)}(\mathbf{q}) = -\frac{2v^2 v_z^2 q_z \mathbf{q}}{[v^2(q_x^2 + q_y^2) + v_z^2 q_z^2]^2}. \quad (7)$$

The superscript “c/v” stands for conduction/valence band. The form of the Berry curvatures resembles the field distribution of a  $z$ -direction point dipole, therefore, this kind of band node is dubbed Berry-dipole node [34,52,53].

A notable property of the dipole Berry curvatures in Eq. (7) is that an integral over a closed surface enclosing the dipole is equal to zero, while an integral over the upper half ( $q_z > 0$ ) or lower half ( $q_z < 0$ ) of the closed surface leads to a nonzero but opposite quantized value [34,51], i.e.,

$$\frac{1}{2\pi} \int_{q_z \gtrless 0} \Omega_{\chi}^{(c)}(\mathbf{q}) \cdot d\mathbf{S} = \mp 1. \quad (8)$$

The above result is a reflection of the fact that the Berry dipole is formed by two opposite and infinitely close Berry charges. From Eq. (7), one can also see that the two low-energy Berry-dipole Hamiltonians lead to the same Berry curvature, which means that the two Berry dipoles are the same. This fact suggests that the net Berry dipole does not need to vanish like the net Berry charge in WSMs.

In WSMs, another well-known characteristic is the open Fermi arcs (isoenergy contour of the surface states’ energy spectrum) connecting the projections of the bulk Weyl nodes in the surface Brillouin zone. A popular picture adopted to understand the origin of Fermi arcs in WSMs is that the Hamiltonians on the 2D momentum planes lying between the Weyl nodes describe Chern insulators with chiral edge states [35]. For the BDSM, we find that the Chern number in each 2D plane between the two Berry-dipole nodes is zero. Nevertheless, we find that, these 2D planes are topologically nontrivial and support a pair of gapless edge states, which form a pair of Fermi arcs on each side surface, as shown in Figs. 2(a)–2(d). While the paired feature of Fermi arcs resembles the scenario in Dirac semimetals (DSMs) [54,55], the two systems are essentially different. In DSMs, a pair of Fermi arcs will carry opposite spin textures as they are related by time-reversal symmetry (TRS). In contrast, here the BDSM lacks TRS, and we find that the pair of Fermi arcs carry the same spin polarization, as shown in Fig. 2(d).

How to understand the counterintuitive existence of a pair of Fermi arcs when the layer Chern number is zero and the TRS is absent? We find that the two-band Hamiltonian at a given  $k_z$  plane between the two Berry-dipole nodes in fact describes a 2D two-band Hopf insulator [56], a class of fixed-band topological insulators beyond the tenfold way classification just similar to its 3D counterpart [36,57–59]. Here we provide an intuitive picture to understand why these 2D layer Hamiltonians with a fixed  $k_z$  are topologically

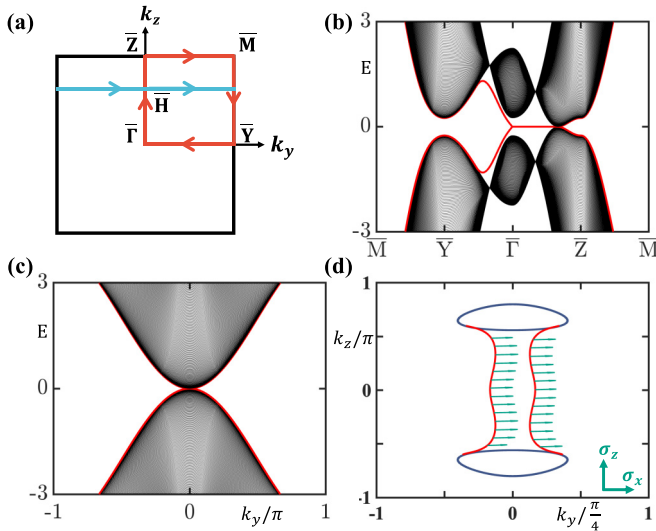


FIG. 2. (a) High symmetry points in the surface Brillouin zone.  $\bar{H}$  refers to the location where one Berry-dipole node is projected. The energy spectra along the two paths in orange and light blue are plotted in (b) and (c). (d) Fermi arcs (red lines) and the projection of bulk Fermi surface (dark blue rings) on the right  $x$ -normal surface. The chemical potential is chosen to be  $\mu = 0.05$ . The green arrows denote the spin polarizations on the Fermi arcs. Common parameters are  $M = 1.5$ ,  $\lambda = t = 1$  and  $\delta = 0$ .

nontrivial and host a pair of gapless edge states. To start, it is instructive to note the following three facts. First, a system is topologically nontrivial as long as it cannot be adiabatically transformed to an atomic insulator (referring to a limit that the Hamiltonian becomes independent of momentum and so all eigenstates exhibit the strongest localization) without the close of bulk or boundary energy gap. Second, the first-order topology can change only when the bulk energy gap gets closed. Third, the bulk energy gap of the BDSM Hamiltonian can only get closed when  $d_x(\mathbf{k})$ ,  $d_y(\mathbf{k})$ , and  $d_z(\mathbf{k})$  simultaneously vanish at some points of the Brillouin zone. These three facts imply that the relative positions of the zero-value contours of  $d_x(\mathbf{k})$ ,  $d_y(\mathbf{k})$ , and  $d_z(\mathbf{k})$  in the Brillouin zone determine whether the Hamiltonian is topologically nontrivial or not [47,60]. Moreover, if any one of  $d_x(\mathbf{k})$ ,  $d_y(\mathbf{k})$ , and  $d_z(\mathbf{k})$  has no zero-value contour, the Hamiltonian must be topologically trivial as it can be adiabatically deformed to the atomic limit. In Fig. 1, we have plotted the contours satisfying  $d_z(\mathbf{k}) = 0$  and the ones simultaneously satisfying  $d_x(\mathbf{k}) = 0$  and  $d_y(\mathbf{k}) = 0$  at the  $k_z = 0$  plane. Apparently, if the latter ones are fixed, the former ones cannot adiabatically be made vanishing without the close of bulk energy gap (more extended discussions are given in Supplemental Material [61]). This suggests that the 2D Hamiltonian at the  $k_z = 0$  plane must be topologically nontrivial as it cannot be adiabatically transformed to an atomic insulator if one fixes two of the three  $d_i(\mathbf{k})$  terms. For convenience, we dub Hamiltonians with such zero-value contour configurations as *conditionally obstructed Hamiltonians*.

Notably, we find that the dimension of the boundary states of these 2D layer Hamiltonians between the two Berry-dipole nodes is the same as the dimension of the zero-value contours

lying between the zero-value contours satisfying  $d_z(\mathbf{k}) = 0$  which cause obstruction. When the dimension is one, the boundary states are 1D propagating edge states; When the dimension is zero, the boundary states are 0D corner states as will be shown in the next section discussing second-order BDSM. As the Chern number is zero, the 1D edge states must show up in pair and be counter-propagating [62], like the scenario in a quantum spin Hall insulator [63–65]. The pair of edge states for the series of 2D layer Hamiltonians between the two Berry-dipole nodes consists of the pair of Fermi arcs of the 3D BDSM.

The existence, stability, and spin polarization of the pair of edge states can further be understood via a 1D topological invariant. The lattice Hamiltonian in Eq. (3) has a  $C_{4z}$  rotation symmetry. Therefore, without loss of generality, we consider the  $k_y = 0$  line in a  $k_z$  plane for illustration. On this high symmetry line, the Hamiltonian reduces as

$$\mathcal{H}(k_x, k_z) = 2\lambda\tilde{M}(k_x, k_z) \sin k_x \sigma_y + [\lambda^2 \sin^2 k_x - \tilde{M}^2(k_x, k_z)] \sigma_z, \quad (9)$$

where  $\tilde{M}(k_x, k_z) = M - t - t \cos k_x - t \cos k_z$ .  $\mathcal{H}(k_x, k_z)$  has chiral symmetry as  $\{\sigma_x, \mathcal{H}(k_x, k_z)\} = 0$ . Therefore, the topological boundary states on the  $x$ -normal surfaces can be characterized by a  $k_z$ -dependent winding number [59],  $W(k_z)$ . Since the two terms are just the real and imaginary part of the complex function,  $[\lambda \sin k_x + i\tilde{M}(k_x, k_z)]^2$ ,  $W(k_z) = 2$  as long as  $\tilde{M}(k_x, k_z) = 0$  has a solution for  $k_x \in (0, \pi)$ .  $W(k_z) = 2$  explains the existence and stability of the edge-state spectrum crossing at  $k_y = 0$  even without TRS, as shown in Fig. 2(b). On the other hand, since the zero-energy boundary states are also the eigenstates of the chiral symmetry operator, the spin polarizations of the two zero-energy states enforced by  $W(k_z) = 2$  on the same  $x$ -normal surface will take the same direction, leading to that the pair of Fermi arcs on a given side surface carries the same spin polarization at the neutrality condition.

**Second-order BDSM.** When  $\delta$  becomes finite, if one only keeps the leading order terms, the low-energy Berry-dipole Hamiltonians retain their forms in Eq. (5). Although the  $\delta$  term has a negligible effect on the low-energy bulk physics, it breaks the  $C_{4z}$  rotation symmetry and the chiral symmetry along high symmetry lines of the  $k_z$  planes. As a result, its effect on the zero-value contour configurations is dramatic, and so as its effect on the boundary states. Once  $\delta \neq 0$ , the ring-shape zero-value contour lying between the two zero-value contours of  $d_z(\mathbf{k})$  immediately becomes four separated points, as shown in Fig. 1(b) (more detailed discussions are given in the Supplemental Material [61]). This kind of 2D conditionally obstructed Hamiltonians correspond to second-order topological insulators/superconductors [47]. Indeed, considering open boundary conditions in  $x$  and  $y$  directions, we find one zero-energy bound state per corner when  $k_z \in \{(\mathbf{k}_{n,-})_z, (\mathbf{k}_{n,+})_z\}$ , as shown in Fig. 3. If one restores the 3D perspective, these zero-energy bound states consist of four hinge Fermi arcs. The above results suggest that the  $\delta$  term will gap out the surface Fermi arcs and lead to the presence of hinge Fermi arcs. In other words, it renders a direct transition from a first-order BDSM to a second-order one.

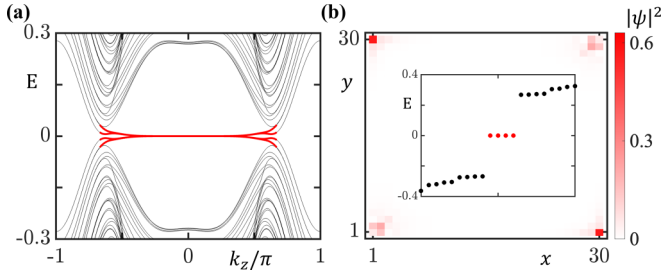


FIG. 3. (a) Energy spectra manifesting second-order topology. Periodic (open) boundary directions in the  $z$  ( $x$  and  $y$ ) direction. (b) The probability density profile of the four zero-energy bound states at  $k_z = 0$ . Parameters are  $M = 1.5$ ,  $\lambda = t = 1$ , and  $\delta = 0.2$ .

The bulk-hinge correspondence in the second-order BDSM shares more similarity with a second-order DSM than a second-order WSM. In a second-order DSM, hinge Fermi arcs can also be obtained by appropriately gapping out the helical surface Fermi arcs [66,67]. In contrast, the hinge Fermi arcs in a second-order WSM cannot directly be descended from the chiral surface Fermi arcs as they are not gappable [68–71]. In addition, if there are only two Weyl nodes in a second-order WSM, the coexistence of surface and hinge Fermi arcs is inevitable.

*Intrinsic anomalous Hall effects and orbital magnetization.* Besides the boundary states, the quadratic energy dispersion and dipole Berry curvature of the low-energy Hamiltonian can give rise to intriguing measurable effects. Here we explore the intrinsic anomalous Hall effects and zero-field orbital magnetization that are directly related to the Berry curvature. In 3D, the intrinsic anomalous Hall conductivity tensor has three independent components, and they are connected to the Berry curvature according to the formula [72] (hereafter we restore the reduced Planck constant)

$$\sigma_{ij} = \frac{e^2}{\hbar} \sum_{\chi} \sum_n \int \frac{d^3q}{(2\pi)^3} \epsilon_{ijk} \Omega_{\chi,k}^{(n)} f(E_{\chi,n}), \quad (10)$$

where  $n$  is the band index and  $f(E_{\chi,n})$  is the Fermi-Dirac distribution function. Considering the zero-temperature limit and a small positive  $\mu$  so that the low-energy Hamiltonian gives a very accurate description, we find  $\sigma_{xz}(\mu) = \sigma_{yz}(\mu) = 0$ , and [61]

$$\sigma_{xy}(\mu) = -\frac{2e^2}{\hbar} \frac{\sqrt{\mu}}{3\pi^2 v_z} = -\frac{2e^2}{h} \frac{D_{F,z}}{3\pi}, \quad (11)$$

where  $D_{F,z} = 2\sqrt{\mu}/v_z$  has the geometric interpretation as the  $z$ -axis diameter of the elliptic Fermi surface. For the two-band Hamiltonian, the orbital magnetic moment is simply related to the Berry curvature [73], i.e.,  $\mathbf{m}_{\chi}(\mathbf{q}) = (e/\hbar)E_{\chi,+}(\mathbf{q})\Omega_{\chi}^{(c)}(\mathbf{q})$ . Notably, we find  $m_{\chi,z}(0, 0, q_z)$  is a constant and can be expressed as [61]

$$m_{\chi,z}(0, 0, q_z) = \frac{-2ev^2}{\hbar} = -2\frac{e\hbar}{2M_{x(y)}^*} = -2\mu_B^*, \quad (12)$$

where  $M_{x(y)}^* \equiv \hbar^2[\partial^2 E_{\chi,+}/\partial q_{x(y)}^2]^{-1} = \hbar^2/2v^2$ , is the  $xy$ -plane effective mass of the conduction-band electrons. We emphasize that this result is different from that of massive Dirac

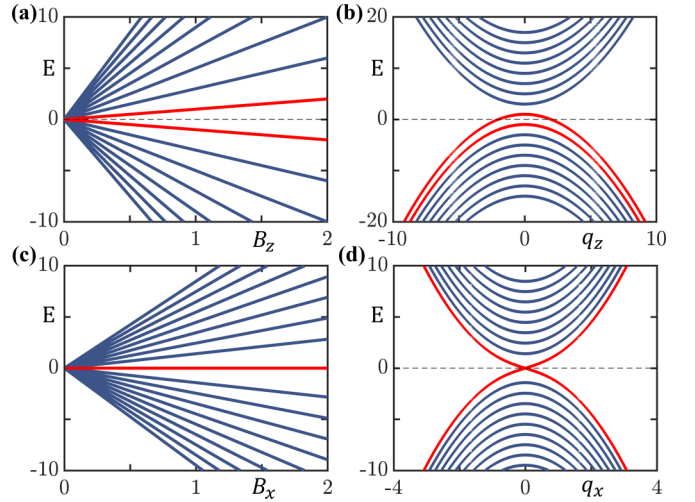


FIG. 4. (a) and (b) Landau levels under a  $z$ -direction magnetic field. [(c), (d)] Landau levels under an  $x$ -direction magnetic field. (a)  $q_z = 0$ , (b)  $B_z = 1$ , (c)  $q_x = 0$ , (d)  $B_x = 1$ . Shared parameters are  $v = 1$  and  $v_z = 0.5$ .  $e = \hbar = 1$  is adopted in all figures.

fermions at the band edge where the factor 2 before the effective Bohr magneton  $\mu_B^*$  is absent [73]. The zero-field orbital magnetization is given by [74]

$$\begin{aligned} \mathbf{M}(\mu) &= \sum_{\chi} \int^{\mu} \frac{d^3q}{(2\pi)^3} \left[ \mathbf{m}_{\chi}(\mathbf{q}) + \frac{e\Omega_{\chi}^{(c)}(\mathbf{q})}{\hbar} (\mu - E_{\chi,+}(\mathbf{q})) \right] \\ &= -\frac{2e}{\hbar} \frac{\mu^{3/2}}{3\pi^2 v_z} \hat{z} = -2n_T \mu_B^* \hat{z} = \frac{\mu}{e} \sigma_{xy}(\mu) \hat{z}, \end{aligned} \quad (13)$$

where  $n = \mu^{3/2}/(3\pi^2 v^2 v_z)$  is the total electron density in the weakly doped regime. The result indicates that all electrons contribute the same magnetization.

*Anisotropic behaviors of Landau levels.* Landau levels (LLs) are of particular interest in the study of topological semimetals as they can reveal the Berry phase of the Fermi surface [75–77]. Since the dipole Berry curvature is quite anisotropic, one can expect that remarkable anisotropic behaviors will show up in the LL structure, even when the zero-field energy spectra are isotropic. Indeed, based on the low-energy Berry-dipole Hamiltonian, we find that, when the magnetic field is along the rotation symmetry axis, i.e.,  $z$  axis, the LLs are analytically solvable and the result is [61]

$$\begin{aligned} E_{\chi,0}(q_z) &= v^2/l_B^2 - v_z^2 q_z^2, \\ E_{\chi,n\pm}(q_z) &= v^2/l_B^2 \pm (2nv^2/l_B^2 + v_z^2 q_z^2), \quad n \geq 1, \end{aligned} \quad (14)$$

where  $l_B = \sqrt{\hbar/eB}$  is the magnetic length and  $n$  is an integer labeling the LLs. A salient feature of the above LLs is that all of them are separated and have a linear dependence on the field strength, as shown in Figs. 4(a) and 4(b). In contrast, when the magnetic field changes to the  $x$  direction, we find that the exact LLs cannot be analytically obtained, but in the small  $q_x$  regime, the LLs take the approximate form [61]

$$\begin{aligned} E_{\chi,1\pm}(q_x) &= \pm \sqrt{2v^3 v_z/l_B^2} q_x, \\ E_{\chi,n\pm}(q_x) &= \pm [F_n v v_z/l_B^2 + G_n v^2 q_x^2], \quad n \geq 2, \end{aligned} \quad (15)$$

where  $F_n = 2\sqrt{n(n-1)}$ ,  $G_n = (2n-1)/2\sqrt{n(n-1)}$ . Compared to the  $z$ -direction case, a remarkable difference is that the two middle LLs cross at  $q_x = 0$ . The numerically determined exact LLs show that the two middle LLs form a 1D nonlinear Dirac cone, as shown in Figs. 4(c) and 4(d).

*Discussions and conclusions.* We have introduced BDSMs into the family of topological semimetals and shown that they carry many salient characteristics both in boundary physics and in bulk physics. This concept can also be generalized to superconducting systems just like other semimetals [78,79]. The bulk responses predicted in our paper are not restricted to our proposed models as they only involve the low-energy Berry-dipole Hamiltonian. Since the low-energy Berry-dipole Hamiltonian can emerge in a number of systems [33,34], an observation of the predicted bulk physics is, in principle,

accessible in real materials, such as mirror-symmetric WSMs with very close Weyl points. The implementation of our models and the observation of the predicted topological boundary states are more suitable to be carried out in artificial systems with high flexibility, such as circuit systems which have recently realized the very related Hopf insulators [42] and acoustic lattices where anisotropic LLs of a three-band Berry-dipole semimetal have been observed [80].

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